

Multiple Choice (10 marks)

1. If a body is cooled below ambient temperature by a system, it can then be used to freeze water. Calculate the maximum air temperature for which freezing takes place.
 - (a) 190
 - (a) 200
 - (a) 170
 - (a) 180
 - (a) 120
2. A Binary number system has how many digits.
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 10
3. The energy stored in the magnetic field in a solenoid 30 cm long and 3 cm diameter wound with 1000 turns of wire carrying a current at 10 amp, is
 - (a) 0.015 joule.
 - (b) 15 joule.
 - (c) 0.0015 joule.
 - (d) 1.5 joule.
 - (e) 0.15 millijoule.
4. If six messages having probabilities $(1/4)$, $(1/4)$, $(1/8)$, $(1/8)$, $(1/8)$ and $(1/8)$ can be accommodated in a single communication system, determine the entropy.
 - (a) 2.00 bits/message
 - (b) 2.25 bits/message
 - (c) 1.75 bits/message
 - (d) 3.00 bits/message
 - (e) 2.75 bits/message
5. Find $f_1(t) \{ _ * \} f_2(t)$ if $f_1(t) = 2 e^{-4t}u(t)$ and $f_2(t) = 5\cos 3tu(t)$.

$$6. f_1(t) \{_{st} \} f_2(t) = (1.6 \cos 3 t + 1.2 \sin 3 t - 1.6 e^{-4t})u(t)$$

$$6. f_1(t) \{_{st} \} f_2(t) = (1.2 \cos 3 t + 1.6 \sin 3 t - 1.3 e^{-4t})u(t)$$

$$6. f_1(t) \{_{st} \} f_2(t) = (1.4 \cos 3 t + 1.2 \sin 3 t - 1.5 e^{-4t})u(t)$$

$$6. f_1(t) \{_{st} \} f_2(t) = (1.6 \cos 3 t + 1.0 \sin 3 t - 1.6 e^{-4t})u(t)$$

True / False (10 marks)

Answer True or False

6. True or False: The attenuation per meter along the waveguide for a wavelength of 2 meters is 545 db/meter.
7. True or False: The z-transform of $f[n] = 3[n+2] + 4[n-1] + 2[n-3]$ is $F(z) = 4 z^2 + 3 z^{-1} + 2z^{-3}$.
7. True or False: The partial-fraction expansion of $F(s)$ is $F(s) = [(1 / 4) / (s + 1)] + [(1 / 2) / (s + 1)^2] - [(1 / 4) / (s + 3)]$.
8. True or False: A box that shows the output results of a control subsystem is referred to as a process box.
9. True or False: The average energy density of a plane wave with a Poynting vector of 5 watts/meter² is (1/4) erg/meter³.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how to calculate the optimal length of a half-wave dipole antenna for a frequency of 300 MHz, considering different velocity factors. Include the implications of these factors on antenna performance.
12. Calculate the number of 300-lumen lamps needed to achieve an average illuminance of 50 lux in a 4 m x 3 m room. Discuss your calculations and the underlying principles of illuminance.

End of Paper